Solutions for Chapter 2

Atmospheric and Oceanic Fluid Dynamics
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2.1 *Show that for an ideal gas in hydrostatic balance, changes in dry static energy ($M = c_p T + gz$) and potential temperature (θ) are related by $\delta M = c_p(T/\theta)\delta\theta$. (The quantity $c_p T/\theta$ is known as the 'Exner function', and is denoted Π.)*

We begin with the expression for potential temperature, $\theta = T(p_R/p)^{R/c_p}$. From this, $\delta\theta = \delta T(p_R/p)^{R/c_p} - RT/(c_p p)(p_R/p)^{R/c_p}\delta p$. Using hydrostasy ($\delta p = \rho g \delta z$) and the ideal gas equation ($p = \rho RT$), this becomes

$$\delta\theta = \delta T\left(\frac{p_R}{p}\right)^{R/c_p} + \frac{g}{c_p}\left(\frac{p_R}{p}\right)^{R/c_p}\delta z = \frac{\theta}{T}\left(\delta T + \frac{g}{c_p}\delta z\right) = \frac{\theta}{Tc_p}\,\delta M, \tag{S2.1}$$

and this is the required result.

2.2 *For an ideal gas in hydrostatic balance, show that:*

(a) The integral of the potential plus internal energy from the surface to the top of the atmosphere [$\int (P + I)\,\mathrm{d}p$] is is equal to its enthalpy;
(b) $\mathrm{d}\sigma/\mathrm{d}z = c_p(T/\theta)\,\mathrm{d}\theta/\mathrm{d}z$, where $\sigma = I + p\alpha + \Phi$ is the dry static energy;
(c) The following expressions for the pressure gradient force are all equal (even without hydrostatic balance):

$$-\frac{1}{\rho}\nabla p = -\theta\,\nabla\Pi = -\frac{c_s^2}{\rho\theta}\nabla(\rho\theta), \tag{S2.2}$$

where $\Pi = c_p T/\theta$ is the Exner function.
(d) Show that item (a) also holds for a gas with an arbitrary equation of state.

(a) The integral is over $z = [0, \infty]$ and equivalently over $p = [p_s, 0]$. Consider the potential energy component, use the ideal gas equation and integrate by parts, as follows:

$$\int_0^{p_s} P\,\mathrm{d}p = \int_0^{p_s} gz\,\mathrm{d}p = \int_0^{\infty} gp\,\mathrm{d}z = \int_0^{p_s} gp\frac{1}{\rho g}\mathrm{d}p = \int_0^{p_s} RT\,\mathrm{d}p \tag{S2.3}$$

The integral of the internal energy is $\int_0^{p_s} c_v T\,\mathrm{d}p$. Adding this to the potential energy integral, and using $c_v + R = c_p$, gives us $\int c_p T\,\mathrm{d}p$. The enthalpy of an ideal gas per unit mass is $c_p T$, so the integral is the total enthalpy of a column of gas of unit area (times a factor of g).

(b) We have $\sigma = I + p\alpha + \Phi = c_v T + RT + gz = c_p T + gz$. Now, from question 2.1, $\delta\sigma = c_p(T/\theta)\delta\theta$ and so $\mathrm{d}\sigma/\mathrm{d}z = c_p(T/\theta)\mathrm{d}\theta/\mathrm{d}z$. (Both σ and M are commony used to denote dry static energy; M is the Montgomery potential.)

(c) Begin with $\theta\nabla\Pi$ and substitute $\Pi = c_p T/\theta = c_p (p/p_s)^{R/c_p}$ and $\theta = T(p_s/p)^{R/c_p}$. After a line of algebra, and using the ideal gas relation, the first result follows.

To obtain the second result, we write $\nabla(\rho\theta) = \nabla(\rho(p_s/p)^{R/c_p} T = \nabla p(P_s/p)^{R/c_p}$. We also write the other θ in terms of p and T, and note that $c_s^2 = \gamma RT$ where $\gamma = c_p/c_v$. After making these subsitutions and a couple of lines of algebra the result follows.

(d) From the fourth expression in (S2.3), the potential energy is just $\int_0^{p_s} p\alpha\,\mathrm{d}p$, and so the potential plus internal energy is $\int_0^{p_s} (I + p\alpha)\mathrm{d}p$. The integrand is just equal to the enthalphy, $h \equiv I + p\alpha$.

2.3 *Show that, without approximation, the unforced, inviscid momentum equation may be written in the forms*

$$\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} = T\nabla\eta - \nabla(p\alpha + I) \tag{S2.4}$$

and

$$\frac{\partial\boldsymbol{v}}{\partial t} + \boldsymbol{\omega}\times\boldsymbol{v} = T\nabla\eta - \nabla B \tag{S2.5}$$

where $\boldsymbol{\omega} = \nabla\times\boldsymbol{v}$, η is the specific entropy ($\mathrm{d}\eta = c_p\,\mathrm{d}\ln\theta$) and $B = I + \boldsymbol{v}^2/2 + p\alpha$ where I is the internal energy per unit mass.

For reversible processes, the first law of thermodynamics may be written as $T\mathrm{d}\eta = \mathrm{d}I + p\mathrm{d}\alpha = \mathrm{d}I + \mathrm{d}(\alpha p) - \alpha\mathrm{d}p$. This implies

$$-\alpha\nabla p = T\nabla\eta - \nabla(I + \alpha p). \tag{S2.6}$$

The inviscid momentum equation may thus be written as

$$\frac{\partial\boldsymbol{v}}{\partial t} + \boldsymbol{v}\cdot\nabla\boldsymbol{v} = T\nabla\eta - \nabla(p\alpha + I) \tag{S2.7}$$

Using the vector identity $\boldsymbol{v}\times(\nabla\times\boldsymbol{v}) = \frac{1}{2}\nabla(\boldsymbol{v}\cdot\boldsymbol{v}) - (\boldsymbol{v}\cdot\nabla)\boldsymbol{v}$, (S2.8) may be written as

$$\frac{\partial\boldsymbol{v}}{\partial t} + \boldsymbol{\omega}\times\boldsymbol{v} = T\nabla\eta - \nabla(p\alpha + I + \frac{1}{2}\boldsymbol{v}\cdot\boldsymbol{v}), \tag{S2.8}$$

which is the form required.

2.4 *Consider two-dimensional fluid flow in a rotating frame of reference on the f-plane. Linearize the equations about a state of rest.*

(a) Ignore the pressure term and determine the general solution to the resulting equations. Show that the speed of fluid parcels is constant. Show that the trajectory of the fluid parcels is a circle with radius $|U|/f$, where $|U|$ is the fluid speed.
(b) What is the period of oscillation of a fluid parcel?
(c) ♦ If parcels travel in straight lines in inertial frames, why is the answer to (b) not the same as the period of rotation of the frame of reference? [To answer this fully you need to understand the dynamics underlying inertial oscillations and inertia circles. See Durran (1993), Egger(1999) and Phillips (2000).]

(a) The equations of motion are

$$\frac{\partial u}{\partial t} - fv = 0, \qquad \frac{\partial v}{\partial t} + fu = 0. \tag{S2.9a,b}$$

Multiplying the first equation by u, the second by v, and adding gives

$$\frac{1}{2}\frac{\partial}{\partial t}(u^2 + v^2) = 0. \tag{S2.10}$$

Hence the speed is constant.

We write the fluid equations as

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{f} \times \boldsymbol{u} = 0, \qquad \text{or} \qquad \frac{\mathrm{d}^2\boldsymbol{x}}{\mathrm{d}t^2} + \boldsymbol{f} \times \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} = 0 \tag{S2.11}$$

where $\boldsymbol{x}$ is the position of a parcel. We can integrate the rightmost expression to give, in component form,

$$\frac{\mathrm{d}x}{\mathrm{d}t} - f(y - y_0) = 0, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} + f(x - x_0) = 0 \tag{S2.12}$$

Writing $X = x - x_0$ and $Y = y - y_0$ gives

$$\frac{\mathrm{d}X}{\mathrm{d}t} - fY = 0, \qquad \frac{\mathrm{d}Y}{\mathrm{d}t} + fX = 0, \tag{S2.13}$$

and these are the equations for motion in a circle — note, for example, that $\mathrm{d}/\mathrm{d}t\,(X^2 + Y^2) = 0$. The period of oscillation is $2\pi/f$, and the velocity of the fluid parcels is $U = (u^2 + v^2)^{1/2}$. The radius of the circle is therefore, by simple geometry, U/f.

(b) From (S2.9) we derive

$$\frac{\partial^2 u}{\partial t^2} + f^2 u = 0. \tag{S2.14}$$

This is the equation for an oscillator with frequency f and period $2\pi/f$. The oscillations are referred to as inertial oscillations, and the trajectory is an inertial circle.

(c) The frequency and period of rotation of the frame of reference are $\Omega = f/2$ and $2\pi/\Omega = 4\pi/f$ respectively, where $\Omega = f/2$. These are not the same as the frequency and period of the inertial oscillations above. That is, the inertial oscillations *do not* correspond to free motion in an inertial reference frame. In the latter case, the motion is completely unforced and the the absolute velocity of the parcels is uniform. In the case of inertial oscillations in a rotating frame, a force of gravity is necessary to keep the parcels in the horizontal surface. For example, if we wished to experimentally demonstrate inertial oscillations in the laboratory, we would have to construct a frictionless rotating table with a surface that sloped upward away from the axis of rotation, in order to counterbalance the centrifugal forces. On the rotating earth the centrifugal force is combined with gravity to give an effective gravity that constrains parcels to geopotential surfaces. That is to say, 'inertial' oscillations are not truly inertial. See the references above for more discussion.

2.5 *A fluid at rest evidently satisfies the hydrostatic relation, which says that the pressure at the surface is given by the weight of the fluid above it. Now consider a* deep *atmosphere on a spherical planet. A unit cross-sectional area at the planet's surface lies beneath a column of fluid whose cross-section increases with height, because the total area of the atmosphere increases with distance away from the centre of the planet. Is the pressure at the surface still given by the hydrostatic relation, or is it greater than this because of the increased mass of fluid in the column? If it is still given by the hydrostatic relation, then the pressure at the surface, integrated* *over the entire area of the planet, is less than the total weight of the fluid; resolve this paradox. But if the pressure at the surface is greater than that implied by hydrostatic balance, explain how the hydrostatic relation fails.*

Hydrostatic balance still holds. The radial component of momentum equation is

$$\frac{\mathrm{D}w}{\mathrm{D}t} - \frac{u^2 + v^2}{r} - 2\Omega u \cos\theta = -\frac{1}{\rho}\frac{\partial p}{\partial r} - g. \tag{S2.15}$$

In a fluid at rest all the terms on the left-hand side vanish and we are unambiguously left with hydrostatic balance.

Thus, indeed, the pressure at the surface integrated over the area of the planet is less than the total weight of the fluid. This is only what is expected: supposed the solid planet were the size of a pea, but had the mass of the Earth, and that the atmosphere were the size of the Earth's atmosphere; it is inconceivable that the weight of the entire atmosphere would be concentrated on the surface area of the pea, for as the pea shrank to zero the pressure force would become infinite. So what 'holds up' the atmosphere? The answer is that the atmosphere holds itself up; a section of the atmosphere is like a self-supporting arch, holding itself up with sideways pressure forces.

2.6 *In a self-gravitating spherical fluid, like a star, hydrostatic balance may be written*

$$\frac{\partial p}{\partial r} = -\frac{GM(r)}{r^2}\rho, \tag{S2.16}$$

where $M(r)$ is the mass interior to a sphere of radius r, and G is a constant. Obtain an expression for the pressure as a function of radius when the fluid (a) has constant density, and (b) is an isothermal ideal gas (if possible). The star is of radius a.

If the density is constant then $M(r) = (4/3)\rho_0 \pi r^3$. Hydrostatic balance is then

$$\frac{\partial p}{\partial r} = -\frac{G4\rho_0\pi r^3}{3r^2}\rho_0 = -\frac{4G\rho_0^2\pi}{3}r. \tag{S2.17}$$

Integrating this equation, and applying a boundary condition that $p = 0$ at $r = a$, gives

$$p = \frac{2G\rho_0^2\pi}{3}(a^2 - r^2). \tag{S2.18}$$

For an isothermal ideal gas the problem is more difficult. We have $\rho = Ap$ where $A = 1/(RT)$ and the mass of a sphere is

$$M(r) = \int_0^r 4\pi r'^2 \rho \, \mathrm{d}r'. \tag{S2.19}$$

Hydrostatic balance, (S2.16), is then

$$\frac{\partial p}{\partial r} = \frac{GA^2 \int_0^r 4\pi r'^2 p(r') \, \mathrm{d}r'}{r^2} p. \tag{S2.20}$$

However, the problem as stated has no solution. The physical difficulty is that at the edge of the star, where pressure is zero, then the density of the gas is also zero, given the (unrealistic) isothermal condition. Thus, the gas provides no weight and therefore the pressure remains zero as we move inwards towards the star's center.

2.7 *At what latitude is the angle between the direction of Newtonian gravity (due solely to the mass of the Earth) and that of effective gravity (Newtonian gravity plus centrifugal terms) the largest? At what latitudes, if any, is this angle zero?*

The centrifugal force, F_c is perpendicular to the axis of rotation, and at the surface of the earth it is given by $F_c = \Omega^2 a \cos\vartheta$ where ϑ is the latitude. The force of Newtonian gravity (i.e., that due solely to the mass of the Earth) is directed toward the center of the Earth (taking the Earth as spherical) and has magnitude g. At the equator the centrifugal force is parallel to Newtonian gravity, and at the pole its magnitude is zero.

In spherical coordinates, the centrifugal force is given by

$$\boldsymbol{F}_c = \Omega^2 a \cos\vartheta \sin\vartheta \mathbf{j} + \Omega^2 a \cos^2\vartheta \mathbf{k} \tag{S2.21}$$

where $\mathbf{j}$ is the unit vector pointing northwards, along a line of longitude, and $\mathbf{k}$ is the unit vector in the radial direction. The Newtonian gravitational force is

$$\boldsymbol{g} = -g\mathbf{k}, \tag{S2.22}$$

where we treat the Earth as spherical. The effective gravity is given by $\boldsymbol{g}_e = \boldsymbol{F}_c + \boldsymbol{g}$. The angle, ϕ, between Newtonian gravity and effective gravity is given by

$$\cos\phi = \frac{(\boldsymbol{F}_c + \boldsymbol{g}) \cdot \boldsymbol{g}}{|\boldsymbol{F}_c + \boldsymbol{g}||\boldsymbol{g}|}, \tag{S2.23}$$

using the definition of the vector dot product. Calculating the angle at which this is a maximum requires quite a lot of algebra, which we do not do here. At the equator $\boldsymbol{F}_c$ and $\boldsymbol{g}$ are aligned, so that $\cos\phi = 1$ and $\phi = 0$. At the pole $\boldsymbol{F}_c = 0$ and so again $\phi = 0$. At intermediate latitudes $\phi \neq 0$.

2.8 ♦ Write the momentum equations in true spherical coordinates, including the centrifugal and gravitational terms. Show that for reasonable values of the wind, the dominant balance in the meridional component of this equation involves a balance between centrifugal and pressure gradient terms. Can this balance be subtracted out of the equations in a sensible way, so leaving a useful horizontal momentum equation that involves the Coriolis and acceleration terms? If so, obtain a closed set of equations for the flow this way. Discuss the pros and cons of this approach versus the geometric approximation discussed in section 2.2.1.

2.9 *For an ideal gas show that the expressions (2.225) and (2.229) are equivalent.*

We begin with the expression, (1.112), relating small changes in potential temperature to small changes in pressure and density, namely

$$\frac{\delta\theta}{\theta} = \frac{1}{\gamma}\frac{\delta p}{p} - \frac{\delta\rho}{\rho}. \tag{S2.24}$$

This implies that vertical derivatives are related by

$$\frac{1}{\theta}\frac{\partial\theta}{\partial z} = \frac{1}{\gamma p}\frac{\partial p}{\partial z} - \frac{1}{\rho}\frac{\partial\rho}{\partial z}. \tag{S2.25}$$

Using the hydrostatic relation, $\partial p/\partial z = -\rho g$ and the ideal gas relation $p = \rho RT$ we obtain

$$\frac{g}{\theta}\frac{\partial\theta}{\partial z} = g\left[-\frac{\rho g}{\gamma p} - \frac{1}{\rho}\frac{\partial\rho}{\partial z}\right] = -g\left[\frac{g}{\gamma RT} + \frac{1}{\rho}\frac{\partial\rho}{\partial z}\right] = -g\left[\frac{g}{c_s^2} + \frac{1}{\rho}\frac{\partial\rho}{\partial z}\right], \tag{S2.26}$$

where, in an ideal gas, $c_s^2 = \gamma p/\rho = \gamma RT$. This demonstrates the equivalence requested.

2.10 *Consider an ocean at rest with known vertical profiles of potential temperature and salinity, $\theta(z)$ and $S(z)$. Suppose we also know the equation of state in the form $\rho = \rho(\theta, S, p)$. Obtain an expression for the buoyancy frequency. Check your expression by substituting the equation of state for an ideal gas and recovering a known expression for the buoyancy frequency.*

The buoyancy frequency is given by, in general

$$N^2 = -\frac{g}{\rho_\theta}\left(\frac{\partial \rho_\theta}{\partial z}\right), \tag{S2.27}$$

where ρ_θ is the locally referenced potential density of the environment. The potential density is given by an equation of state of the form $\rho_\theta = \rho_\theta(\theta, S, p_R)$ where p_R is the reference pressure. Thus,

$$N^2 = -\frac{g}{\rho_\theta}\left(\frac{\partial \rho_\theta}{\partial \theta}\frac{\partial \theta}{\partial z} + \frac{\partial \rho_\theta}{\partial S}\frac{\partial S}{\partial z}\right). \tag{S2.28}$$

For an ideal gas, $S = 0$ and $\rho_\theta = p_R/(R\theta)$. We thus obtain $N^2 = (g/\theta)\partial\theta/\partial z$.

2.11 *Convection and its parameterization*

(a) Consider a Boussinesq system in which the vertical momentum equation is modified by the parameter α to read

$$\alpha^2 \frac{\mathrm{D}w}{\mathrm{D}t} = -\frac{\partial \phi}{\partial z} + b, \tag{S2.29}$$

and the other equations are unchanged. (If $\alpha = 0$ the system is hydrostatic, and if $\alpha = 1$ the system is the original one.) Linearize these equations about a state of rest and of constant stratification (as in section 2.10.1) and obtain the dispersion relation for the system, and plot it for various values of α, including 0 and 1. Show that for $\alpha > 1$ the system approaches its limiting frequency more rapidly than with $\alpha = 1$.

(b) ♦ Argue that if $N^2 < 0$, convection in a system with $\alpha > 1$ generally occurs at a larger scale than with $\alpha = 1$. Show this explicitly by adding some diffusion or friction to the right-hand sides of the equations of motion and obtaining the dispersion relation. You may do this approximately.

(a) The linear equations are, dropping primes,

$$\partial_t u = -\partial_x \phi \tag{S2.30a}$$
$$\alpha^2 \partial_t w = -\partial_z \phi + b \tag{S2.30b}$$
$$\partial_x u + \partial_z w = 0 \tag{S2.30c}$$
$$\partial_t b + wN^2 = 0. \tag{S2.30d}$$

Assume that u, w, ϕ, b all vary as $\exp[\mathrm{i}(kx + mz - \omega t)]$, and substitute in to get

$$-\mathrm{i}\omega u = -\mathrm{i}k\phi \tag{S2.31a}$$
$$-\mathrm{i}\omega\alpha^2 w = -\mathrm{i}m\phi + b \tag{S2.31b}$$

$$iku + imw = 0 \qquad \text{(S2.31c)}$$
$$-\mathrm{i}\omega b + N^2 w = 0. \qquad \text{(S2.31d)}$$

Eliminating all other variables in favor of w (or any other variable for that matter) leads to the dispersion relation

$$\omega^2 = \frac{N^2k^2}{\alpha^2k^2 + m^2}. \qquad \text{(S2.32)}$$

Note that the limiting value of the frequency as $k^2/m^2 \to \infty$ is N/α. We can see just by looking at the dispersion relation that it will approach this limit faster for larger α, though plotting makes it explicit. (Aside: if $\alpha = 0$ we have the hydrostatic limit in which $\omega^2 = N^2k^2/m^2$ and the frequency increases without limit for large horizontal wavenumber. The case with *large* α is sometimes called the *hypohydrostatic* case.)

(b) One way to do this is to add a diffusion term to the thermodynamic term so that

$$\partial_t b + wN^2 = A\nabla^2 b, \qquad -\mathrm{i}\omega b + N^2 w = -Ak^2 b. \qquad \text{(S2.33)}$$

After a little algebra we obtain the dispersion relation:

$$\omega^2(\alpha^2k^2 + m^2) + \mathrm{i}\omega Ak^2(\alpha^2k^2 + m^2) = N^2k^2. \qquad \text{(S2.34)}$$

We can solve this exactly using the formula for solving a quadratic equation, but it is as or more informative just to look at the individual terms. If $N^2 < 0$ there is (even if $A = 0$) an imaginary component that implies an instability, with $\omega \propto \pm\mathrm{i}N$ and so an exponential growth. This is convective instability, as discussed in the text. If $A \neq 0$ the instability is damped by the diffusive term, as this gives a contribution to the frequency of the form

$$\omega = -\mathrm{i}Ak^2, \qquad \text{(S2.35)}$$

and so a time dependance of the form $\exp(-Ak^2t)$. The damping occurs preferentially at high wavenumbers. Thus, if we have both nonzero α and nonzero A, the convective instability will approach its limiting growth rate faster (because of the nonzero α) but the small scales will be damped (by A). Thus, convection may occur at larger scales than in the original system.

2.12 (a) The geopotential height *is the height of a given pressure level. Show that in an atmosphere with a uniform lapse rate (i.e.,* $\mathrm{d}T/\mathrm{d}z = -\Gamma =$ *constant) the geopotential height at a pressure* p *is given by*

$$z = \frac{T_0}{\Gamma}\left[1 - \left(\frac{p_0}{p}\right)^{-R\Gamma/g}\right] \qquad \text{(S2.36)}$$

where T_0 *is the temperature at* $z = 0$.

(b) In an isothermal atmosphere, obtain an expression for the geopotential height as function of pressure, and show that this is consistent with the expression (S2.36) *in the appropriate limit.*

(a) We begin with the hydrostatic relation $\mathrm{d}p = -\rho g\,\mathrm{d}z$. Using the ideal gas equation of state and the given lapse rate we obtain

$$\mathrm{d}p = -\rho g\,\mathrm{d}z = -\frac{pg}{RT}\,\mathrm{d}z = -\frac{pg}{R(T_0 - \Gamma z)}\,\mathrm{d}z \qquad \text{(S2.37)}$$

and therefore

$$\frac{R}{g}\frac{\mathrm{d}p}{p} = \frac{-\,\mathrm{d}z}{T_0 - \Gamma z}. \tag{S2.38}$$

Integrating this gives

$$\left(\frac{p}{p_0}\right)^{R\Gamma/g} = \left(1 - \frac{\Gamma z}{T_0}\right) \tag{S2.39}$$

whence

$$z = \frac{T_0}{\Gamma}\left[1 - \left(\frac{p}{p_0}\right)^{R\Gamma/g}\right]. \tag{S2.40}$$

An interesting special case of this is an isentropic atmosphere ($\theta = \theta_0$) in which $\Gamma = g/c_p$, giving

$$z = \frac{c_p\theta_0}{g}\left[1 - \left(\frac{p}{p_0}\right)^{R/c_p}\right]. \tag{S2.41}$$

The factor $c_p\theta_0/g$ is the total geopotential height of the atmosphere.

(b) We can obtain the corresponding result for an isothermal atmosphere by setting $\Gamma = 0$ in (S2.38), and this gives

$$z = -\frac{RT_0}{g}\ln\frac{p}{p_0}. \tag{S2.42}$$

This gives the familiar result that pressure falls exponentially with height in an isothermal atmosphere, $p = p_0\exp[-gz/(RT_0)]$. We can obtain this result from (S2.41) by taking the limit at $\Gamma \to 0$, but care must be taken as both numerator (the term in square brackets) and denominator (Γ itself) go to zero.

2.13 Show that the inviscid, adiabatic, hydrostatic primitive equations for a compressible fluid conserve a form of energy (kinetic plus potential plus internal), and that the kinetic energy has no contribution from the vertical velocity. You may assume Cartesian geometry and a uniform gravitational field in the vertical direction.

The horizontal and vertical momentum equations in the primitive equations are

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} + \boldsymbol{f}\times\boldsymbol{u} = -\frac{1}{\rho}\nabla_z p, \tag{S2.43}$$

$$0 = -\frac{1}{\rho}\frac{\partial p}{\partial z} - g. \tag{S2.44}$$

where $\boldsymbol{u}$ is the horizontal velocity and ∇_z the horizontal gradient. Take the dot product of the first equation with $\boldsymbol{u}$, and multiply the second equation by w, to obtain the kinetic energy equation

$$\frac{1}{2}\rho\frac{\mathrm{D}\boldsymbol{u}^2}{\mathrm{D}t} = -\nabla\cdot(p\boldsymbol{v}) + p\nabla\cdot\boldsymbol{v} - \rho g w \tag{S2.45}$$

where $\boldsymbol{v}$ is the three-dimensional velocity, and the $\mathrm{D}/\mathrm{D}t$ and ∇ operators are fully three dimensional.

The adiabatic internal energy equation is

$$\rho\frac{\mathrm{D}I}{\mathrm{D}t} = -p\nabla\cdot\boldsymbol{v}, \tag{S2.46}$$

and a 'potential energy equation' may be written as

$$\rho g\frac{\mathrm{D}z}{\mathrm{D}t} = \rho g w. \tag{S2.47}$$

Adding (S2.45), (S2.46) and (S2.47) gives

$$\rho \frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{1}{2}\boldsymbol{u}^2 + I + gz\right) = -\nabla \cdot (p\boldsymbol{v}). \tag{S2.48}$$

Using mass continuity equation in the form $\mathrm{D}\rho/\mathrm{D}t = -\rho\nabla\cdot\boldsymbol{v}$, (S2.48) may be written as

$$\frac{\partial}{\partial t}\left[\rho\left(\frac{1}{2}\boldsymbol{u}^2 + I + gz\right)\right] + \nabla\cdot\left[\rho\boldsymbol{v}\left(\frac{1}{2}\boldsymbol{u}^2 + I + gz + p/\rho\right)\right] = 0, \tag{S2.49}$$

or

$$\frac{\partial E}{\partial t} + \nabla\cdot[\boldsymbol{v}(E+p)] = 0, \tag{S2.50}$$

where $E = \rho(\boldsymbol{u}^2/2 + I + gz)$.

The derivation mirrors the one in the book (section 1.10.2), except for the first step involving the hydrostatic relation instead of the full vertical momentum equation. At a more general level, the hydostatic primitive equations may be regarded as the first term in the expansion in the aspect ratio of the Navier-Stokes equations. The aspect ratio multiplies the vertical velocity and so, in the energy equation, the contribution from the vertical velocity is absent. Similarly, the conserved potential vorticity in the primitive equations does not contain a contribution from the vertical velocity (see section 4.3.5).

2.14 Consider the simple Boussinesq equations, $\mathrm{D}\boldsymbol{v}/\mathrm{D}t = -\nabla\phi + \mathbf{k}b + \nu\nabla^2\boldsymbol{v}$, $\nabla\cdot\boldsymbol{v} = 0$, $\mathrm{D}b/\mathrm{D}t = Q + \kappa\nabla^2 b$. *Obtain an energy equation similar to (2.112), but now with the terms on the right-hand side that represent viscous and diabatic effects. Over a closed volume, show that the dissipation of kinetic energy is balanced by a buoyancy source. Also show that, in a statistically steady state, the heating must occur at a lower level than the cooling if a kinetic-energy dissipating circulation is to be maintained.*

We first obtain a kinetic energy equation by writing the momentum equation in vector-invariant form and taking its dot product with $\boldsymbol{v}$ to give

$$\frac{1}{2}\frac{\partial\boldsymbol{v}^2}{\partial t} = -\nabla\cdot(\boldsymbol{v}B) + wb + \nu\boldsymbol{v}\cdot\nabla^2\boldsymbol{v}, \tag{S2.51}$$

where $B = \boldsymbol{v}^2/2 + \phi$ is the Bernoulli function. Integrating over an enclosed domain gives

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\langle\boldsymbol{v}^2\rangle = \langle wb\rangle - \varepsilon \tag{S2.52}$$

where ε is the energy dissipation rate, $\varepsilon = -\nu < \boldsymbol{v}\cdot\nabla^2\boldsymbol{v} >= \nu < \omega^2 >$. We obtain a potential energy equation from the thermodynamic equation, by writing

$$\frac{\mathrm{D}bz}{\mathrm{D}t} = z\frac{\mathrm{D}b}{\mathrm{D}t} + b\frac{\mathrm{D}z}{\mathrm{D}t} = z\dot{Q} + bw, \tag{S2.53}$$

and integrating this over the domain gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle bz\rangle = \left\langle z\dot{Q}\right\rangle + \langle bw\rangle. \tag{S2.54}$$

The term $\langle bw\rangle$ evidently represents the conversion of potential energy to kinetic energy. Combining (S2.52) with (S2.54) gives the total energy equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\langle\frac{1}{2}\boldsymbol{v}^2 - bz\right\rangle = -\left\langle z\dot{Q}\right\rangle - \varepsilon. \tag{S2.55}$$

and in a statistically steady state we must have $\left\langle z\dot{Q}\right\rangle = -\varepsilon$. The dissipation ε is a positive definite quantity; thus, for there to be a steady state, the quantity $\left\langle z\dot{Q}\right\rangle$ must be negative. That is, the heating must be negatively correlated with height, or the heating must occur at a lower level than the cooling.

Alternate solution from Adam Sobel:
(This solution looks at the dissipative terms in more detail.)

Our equations are

$$\frac{d\boldsymbol{v}}{dt} = -\frac{\nabla p}{\rho_0} + \mathbf{k}b + \nu\nabla^2\boldsymbol{v} \tag{S2.56}$$

$$\frac{db}{dt} = Q + \kappa\nabla^2 b. \tag{S2.57}$$

We will assume here that μ and κ are constants. Following the book, we write

$$\frac{d\Phi}{dt} = -w,$$

and using that together with the buoyancy equation gives

$$\frac{d(b\Phi)}{dt} = -wb + Q\Phi + \kappa\Phi\nabla^2 b. \tag{S2.58}$$

Then we take $\boldsymbol{v}\cdot$ the momentum equation to get the kinetic energy equation,

$$\frac{d(|\boldsymbol{v}|^2/2)}{dt} = bw - \nabla\cdot(\phi v) + \boldsymbol{v}\cdot\nu\nabla^2\boldsymbol{v}. \tag{S2.59}$$

Adding (S2.58) and (S2.59) gives the dissipative analog to (2.112) in Vallis:

$$\frac{\partial}{\partial t}(\frac{1}{2}\boldsymbol{v}^2 + b\Phi) + \nabla\cdot[\boldsymbol{v}(\frac{1}{2}\boldsymbol{v}^2 + b\Phi + \phi)] = Q\Phi + \kappa\Phi\nabla^2 b + \boldsymbol{v}\cdot\nu\nabla^2\boldsymbol{v}. \tag{S2.60}$$

Let's work on the last term, which is the velocity dotted with the viscous force. Let's rewrite things in index notation. The viscous force is the divergence of the stress tensor:

$$\mu\nabla^2\boldsymbol{v} = \nabla\cdot\tau_{ij} = \frac{\partial}{\partial x_j}(2\mu e_{ij}), \tag{S2.61}$$

where we've used Einstein sum notation (sum over any index that appears twice), defined the dynamic viscosity $\mu = \rho_0\nu$, and defined the stress tensor,

$$e_{ij} = \frac{1}{2}(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}).$$

Note that we have used incompressibility in (S2.61) in that

$$\frac{\partial}{\partial x_j}(2e_{ij}) = \frac{\partial}{\partial x_j}(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}) = \frac{\partial^2 v_i}{\partial x_j\partial x_j} + \frac{\partial^2 v_j}{\partial x_j\partial x_i},$$

but

$$\frac{\partial^2 u_j}{\partial x_j\partial x_i} = \frac{\partial}{\partial_i}(\frac{\partial u_j}{\partial x_j}) = 0,$$

since, translating back to vector notation, $\partial v_j/\partial x_j = \nabla\cdot\boldsymbol{v} = 0$. Anyway, getting back on our train of thought, the last term on the RHS of (S2.59) can be written

$$2\mu v_i\frac{\partial}{\partial x_j}(e_{ij}) = 2\mu[\frac{\partial}{\partial x_j}(v_i e_{ij}) - e_{ij}\frac{\partial v_i}{\partial x_j}].$$

Now the so-called velocity gradient tensor, $\partial v_i/\partial x_j$, can be written as the sum of its symmetric and anti-symmetric parts,

$$\frac{\partial v_i}{\partial x_j} = e_{ij} + r_{ij},$$

where r_{ij} is the anti-symmetric part. But the doubly contracted product of a symmetric and anti-symmetric tensor is zero, so

$$e_{ij}\frac{\partial v_i}{\partial x_j} = e_{ij}(e_{ij} + r_{ij}) = e_{ij}e_{ij}.$$

Thus, dividing by ρ_0, we have

$$\boldsymbol{v} \cdot \nu\nabla^2\boldsymbol{v} = 2\nu v_i \frac{\partial}{\partial x_j}(e_{ij}) = 2\nu[\frac{\partial}{\partial x_j}(v_i e_{ij}) - e_{ij}e_{ij}].$$

Now we're getting somewhere. The first term in brackets on the RHS is the divergence of a flux. If we integrate over a closed volume, we can use Gauss' law to convert it to a surface integral, and show that that surface integral vanishes, because it has the velocity in it, and for a motionless boundary in a viscous fluid (requiring a no-slip boundary condition), all three components of the velocity vanish at the boundary.

So, knowing that, let's now integrate (S2.60) over a closed volume. The flux divergence on the left-hand side vanishes too, since it is just velocity times scalars. Let's assume statistical steady state, so we can drop the tendency term too. Then the balance is (in mixed notation)

$$0 = \int(\Phi Q + \kappa\Phi\nabla^2 b)dV - 2\nu\int e_{ij}e_{ij}dV.$$

The first term, containing contributions from the heating as well as diffusion of buoyancy, is the "buoyancy source". The second is the frictional dissipation, which is negative definite since the doubly contracted product $e_{ij}e_{ij}$ is a sum of squares of real quantities. So, we have solved the first part of the problem.

Regarding the second part of the problem. We just have to show that the heating has to occur at a lower level than the cooling in order for a steady state dissipation to be maintained. Let's consider just the true heating term, ΦQ; we'll address the buoyancy diffusion term after. In a sense the result is obvious by inspection; since $\Phi = -z$, we require that on average Q be anti-correlated with z in order to make $\int \Phi Q dV$ positive. For example, just to make things concrete, let's say we have a heating that varies linearly in z,

$$Q = Q_0(0.5 \pm z/z_T), \tag{S2.62}$$

where our domain boundaries are at $z = 0$ and $z = z_T$. Then we have

$$\int \Phi Q dV = Q_0 A \int \Phi Q dz = Q_0 A(z_T^2/4 \pm z_T^2/3),$$

where A is the horizontal area of the domain. The integral will be positive or negative depending whether we choose the plus or minus sign in (S2.62), i.e., whether the heating increases or decreases with height. You can get the same result if you choose a sinusoidal heating, etc.

Now what about the buoyancy diffusion term? Is there some way to reduce that to something with a unique sign, in general? I don't believe so. Just think about it physically. If we have a situation with stable stratification — light fluid on top of heavy fluid — then diffusion of density, as it tries to iron out the density distribution and bring it closer to a state of uniform density, will actually increase the amount of potential energy in the fluid, by making the light fluid at the top heavier and vice versa. Now, if we imagine our buoyancy-driven circulation forced by heating below cooling, in that case it may be possible to show that the stratification on average is unstable, and thus that the buoyancy diffusion is stabilizing (i.e. a potential energy sink), but let's not try to prove that here.

2.15 ♦ Suppose a fluid is contained in a closed container, with insulating sidewalls, and heated from below and cooled from above. The heating and cooling are adjusted so that there is no net energy flux into the fluid. Let us also suppose that any viscous dissipation of kinetic energy is returned as heating, so the total energy of the fluid is exactly constant. Suppose the fluid starts out at rest and at a uniform temperature, and the heating and cooling are then turned on. A very short time afterwards, the fluid is lighter at the bottom and heavier at the top; that is, its potential energy has increased. Where has this energy come from? Discuss this paradox for both a compressible fluid (e.g., an ideal gas) and for a simple Boussinesq fluid.

Contributions are welcome.

2.16 Consider a rapidly rotating (i.e., in near geostrophic balance) Boussinesq fluid on the f-plane.

(a) Show that the pressure divided by the density scales as $\phi \sim fUL$.

(b) Show that the horizontal divergence of the geostrophic wind vanishes. Thus, argue that the scaling $W \sim UH/L$ is an overestimate *for the magnitude of the vertical velocity. (Optional extra: obtain a scaling estimate for the magnitude of vertical velocity in rapidly rotating flow.)*

(c) Using these results, or otherwise, discuss whether hydrostatic balance is more or less likely to hold in a rotating flow than in non-rotating flow.

(a) Geostrophic balance is $\boldsymbol{f} \times \boldsymbol{v} = \nabla\phi$ where $\phi = p/\rho_0$. Then, from a scaling point of view, $p/(\rho_0 L) \sim fU$ or $\phi \sim fUL$.

(b) Geostrophic balance is $\boldsymbol{f} \times \boldsymbol{u} = -\nabla\phi$. Taking the curl of this, and using a vector identity, gives $\boldsymbol{f}\nabla \cdot \boldsymbol{u} = \nabla \times \nabla\phi = 0$. Alternatively, just write $-fv = -\partial\phi/\partial x$ and $fu = -\partial\phi/\partial y$. Cross differentiate to obtain, if f is constant, $f(\partial v/\partial y + \partial u/\partial x) = 0$. Thus, the horizontal divergence of the geostrophic wind vanishes. Note that if f varies we obtain $f(\partial v/\partial y + \partial u/\partial x) + \beta v = 0$, which is the 'geostrophic vorticity equation'.

The mass conservation equation is $\partial w/\partial z = -(\partial u/\partial x + \partial v/\partial y)$. A somewhat naïve scaling might suggest $W/H = U/L$. However, for geostrophic flow the right-hand side vanishes, and in general for small Rossby number flow we can expect some cancellation. Thus, we expect $W < UH/L$. One estimate of W comes by way of the geostrophic vorticity equation, which after using mass continuity gives $\beta v = f\partial w/\partial z$. This suggests the scaling $W = H\beta V/f$. If and only if there are large variations in the Coriolis parameter then $\beta \sim f/L$ and $W = HV/L$.

(c) Hydrostatic balance arises when the vertical acceleration is smaller than the gravitational and pressure gradient terms in the vertical momentum equation, and scaling theory suggests that it is a good approximation when the aspect ratio is small. Intuitively, we expect that if rotation further suppresses vertical motion, hydrostatic balance will be a still better approximation. The analysis of section 2.8.5 (p 91 of Vallis) discusses this in more detail.

2.17 Estimate the size of the zonal wind 5 km above the surface in the mid-latitude atmosphere in summer and winter using (approximate) values for the meridional temperature gradient in the atmosphere. Also estimate the shear corresponding to the pole–equator temperature gradient in the ocean.

[Adam Sobel] We use the thermal wind relation in log-pressure coordinates (2.212), namely

$$f\frac{\partial u_g}{\partial Z} = -\frac{R}{H}\frac{\partial T}{\partial y}, \tag{S2.63}$$

where now Z is log-pressure height, which is fairly close enough to geometric height. Let's use $H = 7\,\mathrm{km}$, $R = 287\,\mathrm{J\,kg^{-1}\,K^{-1}}$, and take the pole-equator temperature difference to be 60 K in winter. The pole-equator distance is approximately 10,000 km, so the mean temperature gradient is $60/10^7 = 6\times10^{-6}\,\mathrm{K\,m^{-1}}$. A midlatitude value for f is $10^{-4}\ s^{-1}$. Plugging in the numbers gives

$$\frac{\partial u_g}{\partial z} \approx 2.5\,\mathrm{m\,s^{-1}\,km^{-1}}, \tag{S2.64}$$

Let us crudely assume there is zero wind at the ground (due to frictional drag there) and that the shear is uniform with height, so that we can just write

$$u(z) \approx \frac{\partial u_g}{\partial z}z. \tag{S2.65}$$

This gives about $12.5\,\mathrm{m\,s^{-1}}$ at 5 km. That is less than what is observed in winter in midlatitudes, because the temperature gradients are not uniformly distributed in latitude but are concentrated in baroclinic zones. You can get a stronger wind by using the local temperature gradient in midlatitudes.

In summer, the pole-equator temperature difference is more like 30 K, so the same calculation gives $u \approx 6\,\mathrm{m\,s^{-1}}$ at 5 km.

In the ocean, using the same pole-equator temperature differences but the Boussinesq thermal wind relation. We neglect salinity and assume a thermal expansion coefficient $\beta_T = 2\times10^{-4}\,\mathrm{K^{-1}}$. Then a difference of 60 K results in a pole-equator density difference of about 1.2%, or a buoyancy difference of that times g, or about $0.1\,\mathrm{m\,s^{-2}}$. Using (2.202) we obtain a shear on the order of $10\,\mathrm{cm\,s^{-1}\,km^{-1}}$. Notice that the problem doesn't ask for a typical current value at some depth... that's because unlike in the atmosphere, where the winds are weakest near the surface, in the ocean the surface has the largest velocities. The surface velocity is the integration constant we need to compute the velocity at some depth, so we can't compute it. Another complication is that the density gradient also drops off with depth relatively fast, also unlike the atmosphere.

2.18 Using approximate but realistic values for the observed stratification, what is the buoyancy period for (a) the mid-latitude troposphere, (b) the stratosphere, (c) the oceanic thermocline, (d) the oceanic abyss?

[Adam Sobel] In the atmosphere, we use the ideal gas expression

$$N^2 = \frac{g}{\theta}\frac{\partial \theta}{\partial z}. \tag{S2.66}$$

At the surface let us take $\theta = T \approx 290\,\mathrm{K}$, while at the tropopause $\theta \approx 350\,\mathrm{K}$. Thus using a 60 K increase over about 10 km to estimate $\partial\theta/\partial z$, taking an intermediate value for θ itself of 320 K, plugging in, and evaluating in terms of the period rather than frequency itself, we obtain $2\pi/N \approx 7.5$ minutes. For the stratosphere, it depends exactly which part of the stratosphere you choose, but using the lowest, isothermal layer we get about 5 minutes.

In the oceanic thermocline, the potential density varies by about $3\,\mathrm{kg\,m^{-3}}$ over about 200 m. Using that to estimate the effective $\partial\rho/\partial z$, and using

$$N = \sqrt{-\frac{g}{\rho_0}\frac{\partial \rho}{\partial z}}, \tag{S2.67}$$

with a mean density of $1000\,\mathrm{J\,kg^{-1}}$, we obtain about 8.5 minutes. Isn't it somewhat interesting that the ocean and atmosphere values are so similar — why should that be? For the abyss, I get a drop of about $0.4\,\mathrm{kg\,m^{-3}}$ over about 4000 m. Plugging that in gives $2\pi/N \approx 105$ minutes, or almost a couple hours. Our parcels will oscillate rather sluggishly down here.

2.19 Consider a dry, hydrostatic, ideal-gas atmosphere whose lapse rate is one of constant potential temperature. What is its vertical extent? That is, at what height does the density vanish? Is this a problem for the anelastic approximation discussed in the text?

The hydrostatic relation is $\partial p/\partial z = -\rho g$, and the ideal gas relation may be written in the form $\rho = p^{1/\gamma} p_R^{\kappa}/(\theta R)$ where $\gamma = c_p/cv$ and $\kappa = R/c_p$. Thus we have

$$\frac{\partial p}{\partial z} = -Ap^{\mu}, \tag{S2.68}$$

where $A = g p_R^{\kappa}/(\theta R)$ and $\mu = c_v/c_p$. There is a similar equation for density. (In the constant temperature case, the equivalent expression is $\partial p/\partial z = -pg/(RT)$, which gives an exponential fall off in pressure.)

Integrating (S2.68) gives

$$p^{\kappa} = -Az\kappa + p_0^{\kappa} \tag{S2.69}$$

where p_0 is a constant, determined by the total mass of the atmosphere. Thus, the pressure (and also the density) fall with height. However, unlike the isothermal case, the pressure will go to zero at a finite height above the atmosphere:

$$z = \frac{p_0^{\kappa}}{A\kappa} = \frac{\theta c_p}{g} \approx 30\,\mathrm{km}, \tag{S2.70}$$

if p_0 is taken the reference pressure for potential temperature and we take $\theta = 300\,\mathrm{K}$. There is nothing particularly inconsistent about this calculation: given the assumption about constant potential temperature, the temperature at any given height in the atmosphere will be larger than the surface value and so its density will be lower than in the isothermal case. Note that the value of z above is the same as the total geopotential height that emerged from question 2.12.

2.20 *Show that for an ideal gas, the expressions (2.229), (2.224) and (2.225) are all equivalent, and express N^2 terms of the temperature lapse rate, $\partial T/\partial z$.*

We begin with the expression, (1.112), relating small changes in potential temperature to small changes in pressure and density, namely

$$\frac{\delta\theta}{\theta} = \frac{1}{\gamma}\frac{\delta p}{p} - \frac{\delta\rho}{\rho}. \tag{S2.71}$$

This implies that vertical derivatives are related by

$$\frac{1}{\theta}\frac{\partial\theta}{\partial z} = \frac{1}{\gamma p}\frac{\partial p}{\partial z} - \frac{1}{\rho}\frac{\partial\rho}{\partial z}. \tag{S2.72}$$

Using the hydrostatic relation, $\partial p/\partial z = -\rho g$ and the ideal gas relation $p = \rho RT$ we obtain

$$\frac{g}{\theta}\frac{\partial\theta}{\partial z} = g\left[-\frac{\rho g}{\gamma p} - \frac{1}{\rho}\frac{\partial\rho}{\partial z}\right] = -g\left[\frac{g}{\gamma RT} + \frac{1}{\rho}\frac{\partial\rho}{\partial z}\right] = -g\left[\frac{g}{c_s^2} + \frac{1}{\rho}\frac{\partial\rho}{\partial z}\right], \tag{S2.73}$$

where, in an ideal gas, $c_s^2 = \gamma p/\rho = \gamma RT$. This demonstrates the equivalence between (2.225) and (2.229).

In an ideal gas potential temperature, θ, and potential density, ρ_θ, are related by $\rho_\theta = p_R/(\theta R)$. Thus

$$\frac{g}{\rho_\theta}\left(\frac{\partial\rho_\theta}{\partial z}\right) = -\frac{g}{\theta}\left(\frac{\partial\theta}{\partial z}\right). \tag{S2.74}$$

This demonstrates the equivalence between (2.224) and (2.225).

From (1.112), variations in temperature, potential temperature and pressure are related by

$$\frac{\delta\theta}{\theta} = \frac{\delta T}{T} - \kappa\frac{\delta p}{p}, \tag{S2.75}$$

and hence vertical derivatives are related by

$$\frac{1}{\theta}\frac{\partial\theta}{\partial z} = \frac{1}{T}\frac{\partial T}{\partial z} - \frac{\kappa}{p}\frac{\partial p}{\partial z} = \frac{1}{T}\frac{\partial T}{\partial z} + \frac{R}{c_p p}\rho g = \frac{1}{T}\left(\frac{\partial T}{\partial z} + \frac{g}{c_p}\right) \tag{S2.76}$$

using the ideal gas equation, $p = \rho RT$, and the hydrostatic relation, $\partial p/\partial z = -\rho g$. Thus, the buoyancy frequency may be written as

$$N^2 = \frac{g}{\widetilde{T}}\left(\frac{\partial\widetilde{T}}{\partial z} + \frac{g}{c_p}\right), \tag{S2.77}$$

with the tilde denoting the environmental values of the variable. A neutral or dry adiabatic lapse rate, $\partial\widetilde{\theta}/\partial z = 0$, therefore corresponds to $\partial\widetilde{T}/\partial z = -g/c_p$.

2.21 ♦ *Calculate an approximate but reasonably accurate expression for the buoyancy equation for seawater. (From notes by R. de Szoeke.)*

Solution (i): the buoyancy frequency is given by

$$N^2 = -\frac{g}{\rho}\left(\frac{\partial\rho_{pot}}{\partial z}\right)_{\text{env}} = \frac{g}{\alpha}\left(\frac{\partial\alpha_{pot}}{\partial z}\right)_{\text{env}} = -\frac{g^2}{\alpha^2}\left(\frac{\partial\alpha_{pot}}{\partial p}\right)_{\text{env}} \tag{S2.78}$$

where $\alpha_{pot} = \alpha(\theta, S, p_R)$ is the potential density, and p_R a reference pressure. From (1.155)

$$\alpha_{pot} = \alpha_0 \left[1 - \frac{\alpha_0}{c_0^2} p_R + \beta_T (1 + \gamma^* p_R)\theta' + \frac{1}{2}\beta_T^* \theta'^2 - \beta_S (S - S_0) \right]. \tag{S2.79}$$

Using this and (S2.78) we obtain the buoyancy frequency,

$$N^2 = -\frac{g^2}{\alpha^2}\alpha_0 \left[\beta_T \left(1 + \gamma p_R + \frac{\beta_T^*}{\beta_T}\theta \right) \left(\frac{\partial \theta}{\partial p} \right)_{\text{env}} - \beta_S \left(\frac{\partial S}{\partial p} \right)_{\text{env}} \right], \tag{S2.80}$$

although we must substitute local pressure for the reference pressure p_R. (Why?)

Solution (ii): the sound speed is given by

$$c_s^{-2} = -\frac{1}{\alpha^2} \left(\frac{\partial \alpha}{\partial p} \right)_{\theta, S} = \frac{1}{\alpha^2} \left(\frac{\alpha_0^2}{c_0^2} - \gamma \alpha_1 \theta \right) \tag{S2.81}$$

and, using (S2.78) and (2.229) the square of the buoyancy frequency may be written

$$N^2 = \frac{g}{\alpha} \left(\frac{\partial \alpha}{\partial z} \right)_{\text{env}} - \frac{g^2}{c_s^2} = -\frac{g^2}{\alpha^2} \left[\left(\frac{\partial \alpha}{\partial p} \right)_{\text{env}} + \frac{\alpha^2}{c_s^2} \right]. \tag{S2.82}$$

Using (1.155), (S2.81) and (S2.82) we recover (S2.80), although now with p explicitly in place of p_R.

2.22 (a) Use the chain rule to show that the horizontal gradients of a field in height coordinates and in ξ coordinates are related by

$$\nabla_z \Psi = \nabla_\xi \Psi - (\partial \Psi / \partial \xi)(\partial \xi / \partial z) \nabla_\xi z. \tag{S2.83}$$

(b) Show that w, the vertical velocity in height coordinates, may be expressed in ξ coordinates as

$$w = \mathrm{D}z/\mathrm{D}t = (\partial z / \partial t)_\xi + \boldsymbol{u} \cdot \nabla_\xi z + \dot{\xi} \partial z / \partial \xi. \tag{S2.84}$$

(c) Use the above expressions to verify (2.143), the expression for the material derivative in ξ coordinates.

2.23 Begin with the mass conservation in height coordinates, namely $\mathrm{D}\rho/\mathrm{D}t + \rho \nabla \cdot \boldsymbol{v} = 0$. Transform this into pressure coordinates using the chain rule (or otherwise) and derive the mass conservation equation in the form $\nabla_p \cdot \boldsymbol{u} + \partial \omega / \partial p = 0$.

2.24 ♦ Starting with the primitive equations in pressure coordinates, derive the form of the primitive equations of motion in sigma-pressure coordinates. In particular, show that the prognostic equation for surface pressure is,

$$\frac{\partial p_s}{\partial t} + \nabla \cdot (p_s \boldsymbol{u}) + p_s \frac{\partial \dot{\sigma}}{\partial \sigma} = 0 \tag{S2.85}$$

and that hydrostatic balance may be written $\partial \Phi / \partial \sigma = -RT/\sigma$.

2.25 Starting with the primitive equations in pressure coordinates, derive the form of the primitive equations of motion in log-pressure coordinates in which $Z = -H \ln(p/p_r)$ is the vertical coordinate. Here, H is a reference height (e.g., a scale height RT_r/g

where T_r is a typical or an average temperature) and p_r is a reference pressure (e.g., 1000 mb). In particular, show that if the 'vertical velocity' is $W = \mathrm{D}Z/\mathrm{D}t$ then $W = -H\omega/p$ and that

$$\frac{\partial \omega}{\partial p} = -\frac{\partial}{\partial p}\left(\frac{pW}{H}\right) = \frac{\partial W}{\partial Z} - \frac{W}{H}. \tag{S2.86}$$

*and obtain the mass conservation equation (**??**). Show that this can be written in the form*

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{1}{\rho_s}\frac{\partial}{\partial Z}(\rho_s W) = 0, \tag{S2.87}$$

where $\rho_s = \rho_r \exp(-Z/H)$.

2.26 *(a) Prove that the argument of the square root in (2.260) is always positive.*

(b) ♦ This argument seems to depend on the parameters in the ideal gas equation of state. Is it more general than this? Is a natural system possible for which the argument is negative, and if so what physical interpretation could one ascribe to the situation?

(a) The largest value or the argument occurs when $m = 0$ and $k^2 = 1/(4H^2)$. The argument is then $1 - 4H^2N^2/c_s^2$. But $c_s^2 = \gamma R T_0 = \gamma g H$ and $N^2 = g\kappa/H$ so that $4N^2H^2/c_s^2 = 4\kappa/\gamma \approx 0.8$.

(b) Contributions from readers are welcome.

2.27 *Consider a wind stress imposed by a mesoscale cyclonic storm (in the atmosphere) given by*

$$\boldsymbol{\tau} = -A\mathrm{e}^{-(r/\lambda)^2}(y\,\mathbf{i} - x\,\mathbf{j}) \tag{S2.88}$$

where $r^2 = x^2 + y^2$, and A and λ are constants. Also assume constant Coriolis gradient $\beta = \partial f/\partial y$ and constant ocean depth H. In the ocean, find (a) the Ekman transport, (b) the vertical velocity $w_E(x, y, z)$ below the Ekman layer, (c) the northward velocity $v(x, y, z)$ below the Ekman layer and (d) indicate how you would find the westward velocity $u(x, y, z)$ below the Ekman layer.

(a) The mass transport is given by $\boldsymbol{M}_E = -\mathbf{k} \times \boldsymbol{\tau}/f$ and substituting the above wind stress gives

$$\boldsymbol{M}_E = \frac{A}{f}\exp(r/\lambda)^2(x\mathbf{i} + y\mathbf{j}). \tag{S2.89}$$

(b) The vertical velocity at the edge of the Ekman layer is equal to the divergence of the velocity in the Ekman layer. Thus,

$$\begin{aligned} w_B &= \frac{1}{\rho_0}\nabla\cdot\boldsymbol{M}_E = \frac{1}{\rho_0}\mathbf{k}\cdot\nabla\times\frac{\boldsymbol{\tau}}{f} \\ &= \frac{A}{f\rho_0}\left[\frac{\partial}{\partial x}\left(\mathrm{e}^{-(x^2+y^2)/\lambda^2}x\right) + \frac{\partial}{\partial y}\left(\mathrm{e}^{-(x^2+y^2)/\lambda^2}y\right)\right] - \frac{A\beta y}{f^2}e^{-(r/\lambda)^2} \\ &= \frac{2A}{f\rho_0}\mathrm{e}^{-r^2/\lambda^2}\left[1 - \frac{r^2}{\lambda^2} - \frac{\beta y}{f}\right]. \end{aligned} \tag{S2.90}$$

This is nonzero, even when $\beta = 0$. Note that the cyclonic flow in the atmosphere has zero divergence, but it causes a divergent flow in the oceanic Ekman layer, and so a vertical velocity at the base of the Ekman layer.

(c) The northward velocity below the Ekman layer is given using the linear vorticity balance (which is not properly encountered until a later chapter); that is

$$\beta v = f\frac{\partial w}{\partial z}. \tag{S2.91}$$

If we integrate over the depth of the fluid below the Ekman layer, and suppose that $w = 0$ at the bottom of the ocean, then

$$\beta[v] = f w_B = \frac{f}{\rho_0}\,\mathrm{curl}_z\frac{\boldsymbol{\tau}}{f}, \tag{S2.92}$$

where $[v]$ is the vertically integrated vertical velocity over the ocean interior, w_B is given by (S2.90), and $\mathrm{curl}_z \equiv \mathbf{k}\cdot\nabla\times$. The above expression differs slightly from 'Sverdrup balance', in which the vertical integral of the entire water column, which we denote as V, is given by the wind-stress at the surface; thus

$$\beta V = \frac{1}{\rho_0}\mathrm{curl}_z\boldsymbol{\tau}. \tag{S2.93}$$

There is only a difference in the two expressions when β is non-zero. The difference between the two expressions is just the transport in the Ekman layer itself, and it may be readily verified that this is given by $V_E = \tau^x/f$ where $V_E = \int \rho_0 v\,dz$ where the integral is over the Ekman layer. This is, of course, just the polewards Ekman transport, essentially by definition.

(d) To obtain the zonal transport beneath the Ekman layer requires a more complete solution to the ocean circulation problem. The zonal transport in the Ekman layer, U_E, is given by $V_E = -\tau^y/f$. The vertically integrated transport over the entire water column is given by Sverdrup balance, (S2.93), which can be written as $\beta\partial\Psi/\partial x = \mathrm{curl}_z\boldsymbol{\tau}$, where Ψ is a mass-transport streamfunction, so that

$$\Psi(x,y) = \int_x^{x_E}\mathrm{curl}_z\boldsymbol{\tau}(x',y)\,\mathrm{d}x' \tag{S2.94}$$

where x_E is the eastern boundary. The zonal velocity is the y-derivative of this. Chapter 14 discusses this type of problem in more detail.

2.28 ♦ *In an atmospheric Ekman layer on the f-plane let us write the momentum equation as*

$$\boldsymbol{f}\times\boldsymbol{u} = -\nabla\phi + \frac{1}{\rho_a}\frac{\partial\boldsymbol{\tau}}{\partial z}, \tag{S2.95}$$

where $\boldsymbol{\tau} = A\rho_a\partial\boldsymbol{u}/\partial z$ and A is a constant eddy viscosity coefficient. An independent *formula for the stress at the ground is $\boldsymbol{\tau} = C\rho_a\boldsymbol{u}$, where C is a constant. Let us take $\rho_a = 1$, and assume that in the free atmosphere the wind is geostrophic and zonal, with $\boldsymbol{u}_g = U\mathbf{i}$.*

(a) Find an expression for the wind vector at the ground. Discuss the limits $C = 0$ and $C = \infty$. Show that when $C = 0$ the frictionally-induced vertical velocity at the top of the Ekman layer is zero.

(b) Find the vertically integrated horizontal mass flux caused by the boundary layer.

(c) When the stress on the atmosphere is $\boldsymbol{\tau}$, the stress on the ocean beneath is also $\boldsymbol{\tau}$. Why? Show how this consistent with Newton's third law.

(d) *Determine the direction and strength of the surface current, and the mass flux in the oceanic Ekman layer, in terms of the geostrophic wind in the atmosphere, the oceanic Ekman depth and the ratio ρ_a/ρ_o, where ρ_o is the density of the sea-water. Include a figure showing the directions of the various winds and currents. How does the boundary-layer mass flux in the ocean compare to that in the atmosphere? (Assume, as needed, that the stress in the ocean may be parameterized with an eddy viscosity.)*

(a) Although we may obtain a solution using the same method that used to obtain (2.30), a useful trick in Ekman layer problems is to write the velocity as a complex number, $\hat{u} = u + \mathrm{i}v$ and $\hat{u}_g = u_g + \mathrm{i}v_g$. The Ekman layer equation, (2.297a), may then be written as

$$A\frac{\partial^2 \hat{U}_a}{\partial z^2} = \mathrm{i}f\hat{U}, \tag{S2.96}$$

where $\hat{U}_a = \hat{u} - \hat{u}_g$. The solution to this is

$$\hat{u} - \hat{u}_g = [\hat{u}_s - \hat{u}_g]\exp\left[-\frac{(1+\mathrm{i})z}{d}\right], \tag{S2.97}$$

where $d = \sqrt{2A/f}$, $\hat{u}_g = U$, $\hat{u}_s = \hat{u}(0)$, and the boundary condition of finiteness at infinity eliminates the exponentially growing solution. The boundary condition at $z = 0$ is $\partial\hat{u}/\partial z = (C/A)\hat{u}$; applying this gives

$$[\hat{u}(0) - \hat{u}_g]\exp(\mathrm{i}\pi/4) = -Cd\hat{u}(0)/(\sqrt{2}A) \tag{S2.98}$$

from which we obtain $\hat{u}_s$. Specifically,

$$\hat{u}_s = U\left[1 + Cd/(\sqrt{2}A)\exp(-\mathrm{i}\pi/4)\right]^{-1} \tag{S2.99}$$

and so

$$u_s = U\frac{1+\gamma}{(1+\gamma)^2+\gamma^2}, \qquad v_s = U\frac{\gamma}{(1+\gamma)^2+\gamma^2}. \tag{S2.100}$$

where $\gamma \equiv Cd/(2A)$.

In the limit $C = 0$ then $\gamma = 0$ and $\boldsymbol{u}_s = U\mathbf{i}$ and $v_s = 0$; this is the free slip case, with no stress at the surface. In this case, there is no Ekman layer and the surface flow is just the interior geostrophic flow. In the limit $C = \infty$, then $\gamma = \infty$ and $\boldsymbol{u}_s = 0$; this is the no slip case. There is, however, a non -zero surface stress.

(b) The mass flux, $\boldsymbol{M}$, or, in complex notation, $\widehat{M}$, is given by

$$\begin{aligned}\widehat{M} &= \int_0^\infty (\hat{u} - \hat{u}_g)\rho_a\,\mathrm{d}z = \int_0^\infty [\hat{u}_s - \hat{u}_g]\exp\left[-\frac{(1+\mathrm{i})z}{d}\right]\rho_a\,\mathrm{d}z \\ &= \frac{1}{2}[\hat{u}_s - \hat{u}_g]\rho_a d(1-\mathrm{i}).\end{aligned} \tag{S2.101}$$

In the free-slip case, the mass flux is zero (there is no boundary layer). In the no-slip case we use (S2.101) and (S2.100), and after a little algebra the components of $\boldsymbol{M}$ are found to be:

$$M^x = -\rho_a U d\frac{\gamma^2}{(1+\gamma)^2+\gamma^2}, \qquad M^y = \rho_a U d\frac{\gamma(1+\gamma)}{(1+\gamma)^2+\gamma^2}. \tag{S2.102}$$

(c) If we consider the ocean and atmosphere to be a single fluid with a change in density, then the stress must be continuous, otherwise there would be infinite accelerations, as the force on the fluid is proportional to $\partial \boldsymbol{\tau}/\partial z$.

(d) Regarding the currents etc. Given an oceanic eddy viscosity A_o, the solution follows the development of section 2.12.4. Specifically, with zero interior ocean currents, the ocean Ekman layer currents are given by

$$u = \frac{\sqrt{2}}{\rho_o f d_o} \mathrm{e}^{z/d} \left[\tau^x \cos(z/d - \pi/4) - \tau^y \sin(z/d - \pi/4)\right], \tag{S2.103}$$

and

$$v = \frac{\sqrt{2}}{\rho_o f d_o} \mathrm{e}^{z/d} \left[\tau^x \sin(z/d - \pi/4) + \tau^y \cos(z/d - \pi/4)\right]. \tag{S2.104}$$

where $\tau^x = C\rho_a u_s$ and $\tau^y = C\rho_a v_s$, with u_s and v_s given by (S2.100). The subscript o denotes an oceanic value, and in particular $d_o = \sqrt{2A_o/f}$ where A_o is the oceanic eddy viscosity. At the surface, we have [c.f. (2.315)]

$$u_{so} = \frac{1}{\rho_o f d_o}(\tau^x + \tau^y), \qquad v_{so} = -\frac{1}{\rho_o f d_o}(\tau^x - \tau^y). \tag{S2.105}$$

Using (S2.100) we obtain

$$u_{so} = U \frac{C\rho_a}{\rho_o f d_o} \frac{1 + 2\gamma}{(1+\gamma)^2 + \gamma^2}, \qquad v_{so} = -U \frac{C\rho_a}{\rho_o f d_o} \frac{1}{(1+\gamma)^2 + \gamma^2}. \tag{S2.106}$$

It is informative to express the surface current in terms of the atmospheric eddy diffusivity, rather than the drag coefficient, using $A \partial \boldsymbol{u}/\partial z = C\boldsymbol{u}$ at the surface. At the surface, from (S2.98), we have

$$\hat{\tau}_s = C\rho_a \hat{u}_s = -\frac{A\rho_a}{d}(1 + \mathrm{i})(\hat{u}_s - \hat{u}_g). \tag{S2.107}$$

Using this expression in (S2.105), and being careful to take the real and imaginary parts first, we obtain, after some algebra,

$$u_{so} = \frac{\rho_a}{\rho_o}\sqrt{\frac{A}{A_o}}\, U \frac{2\gamma^2 + \gamma}{(1+\gamma)^2 + \gamma^2}, \qquad v_{so} = -\frac{\rho_a}{\rho_o}\sqrt{\frac{A}{A_o}}\, U \frac{2\gamma}{(1+\gamma)^2 + \gamma^2}. \tag{S2.108}$$

In the free-slip case ($\gamma = 0$) there is a surface wind but no stress, and the induced ocean surface current is zero. In the no-slip case ($\gamma \to \infty$), (S2.108) gives

$$u_{so} = -\frac{\rho_a}{\rho_o}\frac{A}{A_o} U, \qquad v_{so} = 0. \tag{S2.109}$$

That is, in the no-slip case, the wind stress induces an ocean current *parallel* to the geostrophic current in the atmosphere. The ocean current is smaller than the atmospheric current by an amount equal to the ratio of the densities of the two media, and the ratio of the square root of the kinematic eddy viscosities. This result can be derived more simply, if approximately, by using the fact that stress is continuous across the interface. Thus, at the interface,

$$A_a \rho_a \frac{\partial \boldsymbol{u}_a}{\partial z} = A_o \rho_o \frac{\partial \boldsymbol{u}_o}{\partial z} \tag{S2.110}$$

implying $A_a \rho_a u_a / d_a \sim A_o \rho_o u_o d_o$, explicitly using the subscript a to denote the atmosphere. The depths of the respective Ekman layers are given by $(d_a, d_o) = \sqrt{2(A, A_o)/f}$; thus

$$\sqrt{A_a} \rho_a u_a \sim \sqrt{A_o} \rho_o u_o. \tag{S2.111}$$

For the atmosphere, observations suggest A_a might vary between 10^1 and $10^3\,\mathrm{m^2\,s^{-1}}$ or more near the surface (and see also the next question). For the upper ocean, estimates of A_o vary between 10^{-3} and $10^{-1}\,\mathrm{m^2\,s^{-1}}$, so that using (S2.111) we might expect the ocean surface velocity to be less than that of the atmosphere by a factor of ten or a hundred (it is the ratio of the densities that makes the ocean velocity less). Neverthess, the *mass flux* in the two Ekman layers is equal and opposite, regardless of the values of the eddy viscosities.

Regarding the boundary layer mass flux: The stress at the top of the ocean is equal to the stress at the bottom of the atmosphere. Thus, using (2.290), the mass flux in the ocean is equal and opposite to that in the atmosphere. This is *independent* of the detailed form of the frictional parameterization in the two Ekman layers. Specifically, the mass fluxes are the negative of (S2.102). Equivalently, and from first principles, if we integrate frictional geostrophic balance from the bottom of the ocean Ekman layer (where the stress is zero) to the top of the atmospheric Ekman layer (where the stress is also zero) we see that the net frictional mass transport is zero, so the mass transport in the ocean is equal and opposite to that in the atmosphere. The keen student can evaluate the oceanic mass flux explicitly to check this if s/he wishes.

N. B. The algebra can cause problems in this exercise, so it is important to have a sense of what is physically sensible.

2.29 *The logarithmic boundary layer*

Close to ground rotational effects are unimportant and small-scale turbulence generates a mixed layer. *In this layer, assume that the stress is constant and that it can be parameterized by an eddy diffusivity the size of which is proportional to the distance from the surface. Show that the velocity then varies logarithmically with height.*

Write the stress as $\tau = \rho_0 u^{*2}$ where the constant u^* is called the friction velocity. Using the eddy diffusivity hypothesis this stress is given by

$$\tau = \rho_0 u^{*2} = \rho_0 A \frac{\partial u}{\partial z} \quad \text{where} \quad A = u^* k z, \tag{S2.112}$$

where k is von Karman's ('universal') constant (approximately equal to 0.4). From (S2.112) we have $\partial u/\partial z = u^*/(kz)$ which integrates to give $u = (u^*/k)\ln(z/z_0)$. The parameter z_0 is known as the roughness length. (This is typically of order centimetres or meters depending on the surface.)